



CHAPTER 19 Catalysts

As you have seen in Chapter 18, chemical reactions occur at many different rates. The explosion of gunpowder and combustion of petrol in a car's engine occur very quickly. On the other hand, weathering of buildings, ripening of fruit and rusting of iron all occur quite slowly.

In chemical industry, the rate of a chemical reaction must be high enough to make products at an economical rate. Catalysts increase the rate of a chemical reaction that might otherwise be uneconomical. As well, nearly all reactions in living organisms depend on biological catalysts called enzymes.

By the end of this chapter, you will understand what a catalyst is and their role in changing the reaction rate in relation to reaction pathways and the energy changes occurring during reactions.

Using collision theory, you will learn the effect of a catalyst on the rate of chemical reactions, and explain these effects.

Science as a Human Endeavour

- Catalysts are used in many industrial processes in order to increase the rates of reactions that would otherwise be uneconomically slow. Catalysts are also used to reduce the emission of pollutants produced by car engines. Motor vehicles have catalytic converters which are used to catalyse reactions that reduce the amount of carbon monoxide, unburnt petrol and nitrogen oxides that are emitted.

Science understanding

- catalysts, including enzymes and metal nanoparticles, affect the rate of certain reactions by providing an alternative reaction pathway with a reduced activation energy, hence increasing the proportion of collisions that lead to a chemical change

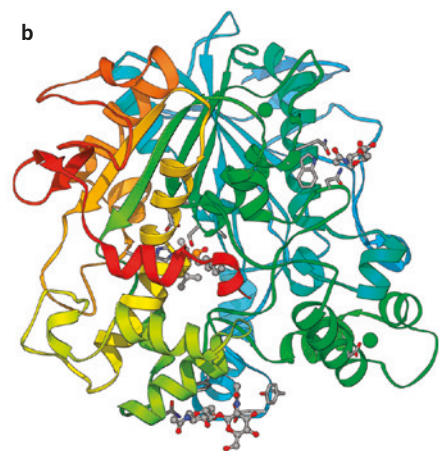
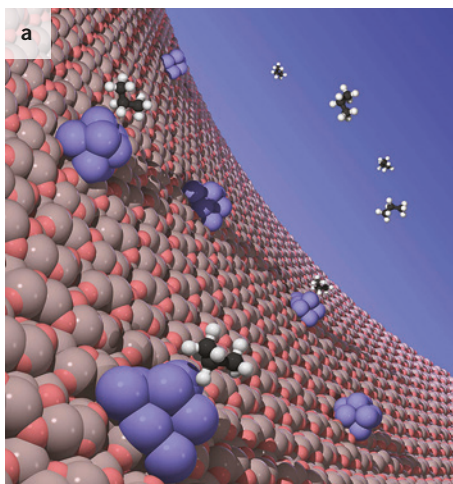


FIGURE 19.1.1 (a) A representation of the structure of a synthetic zeolite catalyst that is widely used in petroleum refineries to break down large hydrocarbon molecules into smaller, more useful molecules. Zeolite is a silica–alumina mineral. (b) A model of a lipase enzyme. Lipase is a catalyst that breaks down fats in the digestive system of the human body.

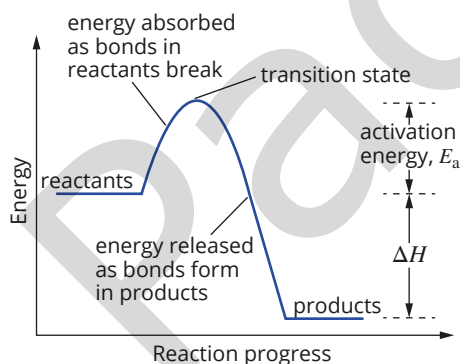


FIGURE 19.1.2 Energy profile diagram of an exothermic reaction such as burning natural gas

19.1 Catalysts

In this section, you will learn how catalysts increase the rate of reaction.

The change in reaction rate when a catalyst is present is often very substantial. The action of a catalyst can be understood using collision theory and the changes in energy that occur during a chemical reaction. Catalysts play an important role in industrial chemistry, in limiting air pollution and in controlling biochemical processes, as shown in the examples in Figure 19.1.1.

CATALYSTS AND ACTIVATION ENERGY

You have already learnt that the potential energy changes associated with a reaction can be represented as an energy profile diagram of the reaction. An energy profile diagram for an exothermic reaction is shown in Figure 19.1.2.

The activation energy is the minimum amount of energy required for a reaction to take place. On the energy profile diagram, the activation energy is measured from the energy of the reactants to the peak of the energy profile diagram.

The enthalpy change, ΔH , can also be represented on an energy profile diagram and is equal to the difference in energy between the products and the reactants.

Some reactions occur very readily because they have very small activation energies, E_a . These reactions need only a small amount of energy to be absorbed for bonds in the reactants to be broken.

Many reactions occur much more rapidly in the presence of a particular element or compound. These substances are known as catalysts. Catalysts are not consumed during the reactions they speed up, and so do not appear as either reactants or products in reaction equations.

Catalysts are able to cause a reaction to occur more quickly because they provide an alternative **reaction pathway**, which causes the activation energy barrier of the overall reaction to be dramatically reduced, as seen in the energy profile diagram in Figure 19.1.3. It is important to note that the original higher energy reaction pathway is still available but with the catalyst present a second, lower energy pathway is also available to the reactants.

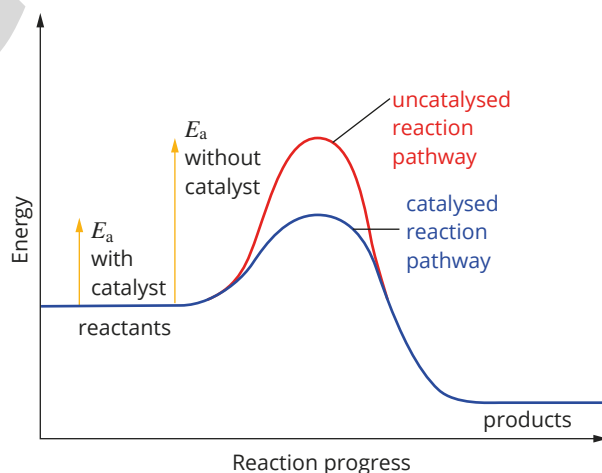


FIGURE 19.1.3 Energy profile diagrams of a catalysed and an uncatalysed reaction

With the catalyst present and a lower activation energy, the colliding particles are more likely to have energies that exceed this lower barrier, causing the bonds in the reactants to be broken more frequently. As a result, a greater proportion of collisions are ‘successful’; that is, they lead to the formation of products. Thus, the reaction rate is increased.

The Maxwell–Boltzmann curve in Figure 19.1.4 shows the smaller number of particles with energies that exceed the activation energy in an uncatalysed reaction (shaded in red) compared with the number of particles with energies that exceed the activation energy in a catalysed reaction (regions shaded in red and blue).

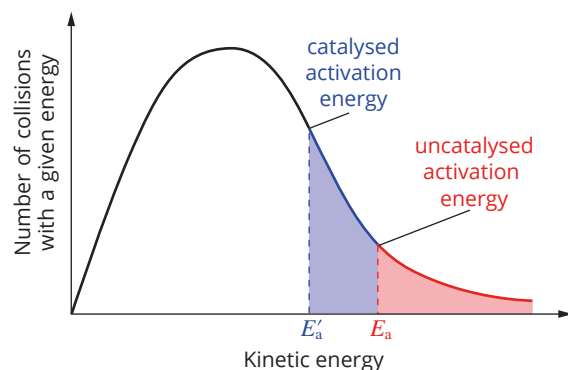


FIGURE 19.1.4 A catalyst provides a reaction pathway with a lower activation energy, increasing the proportion of collisions that exceed the activation energy and lead to a reaction.

i Catalysts lower the activation energy by providing an alternative reaction pathway for the reaction. This increases the proportion of reactant particles with energies greater than the lower activation energy. This increases the proportion of successful collisions.

CHEMFILE

Routes from Melbourne to Canberra

Figure 19.1.5 shows two groups of tourists travelling from Melbourne to Canberra by two different routes. The slower route can be likened to the progress of an uncatalysed reaction; the faster route can be likened to the progress of a catalysed reaction.

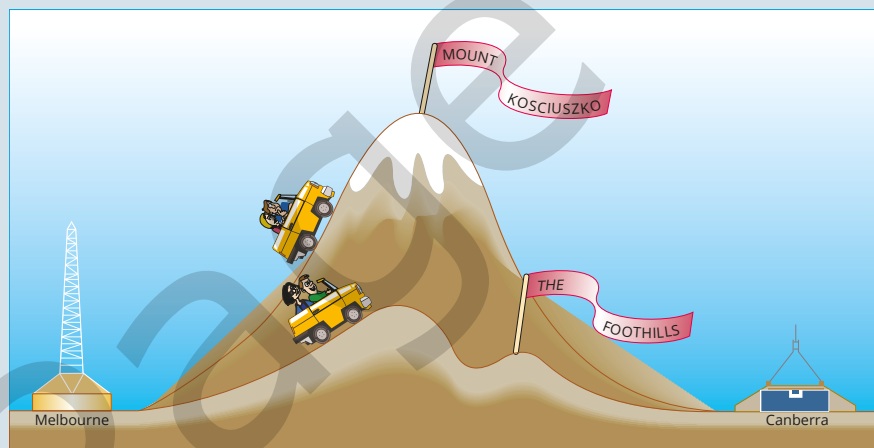


FIGURE 19.1.5 An analogy for the role of a catalyst in a chemical reaction

TYPES OF CATALYSTS

Depending on the physical state of the chemicals involved in the reaction and the catalyst, catalysts can be divided into two groups.

- **Homogeneous catalysts** are in the same physical state as the reactants and products of the reaction.
- **Heterogeneous catalysts** are in a different physical state from the reactants and products of the reaction.

An example of homogeneous **catalysis** occurs in the upper atmosphere and has contributed to the depletion of the ozone layer. Chlorine atoms in the gaseous state act as catalysts in the decomposition of ozone gas into oxygen gas. The chlorine atoms may have come from chlorofluorocarbons (CFCs) released into the atmosphere from refrigerators or air conditioners.

You may be familiar with the catalysed decomposition of a hydrogen peroxide solution using the black powder manganese(IV) oxide (MnO_2). This is an example of the use of a heterogeneous catalyst.

i **Catalysis** is the increase in the rate of a chemical reaction due to the presence of a catalyst.

CATALYSTS IN INDUSTRY

The chemical industry uses catalysts extensively. Chemists prefer to use heterogeneous catalysts for industrial processes because they are:

- more easily separated from the products of a reaction
- much easier to reuse
- able to be used at high temperatures.

Particles at the surface of some solids of high surface area tend to adsorb (form a bond with) gas molecules that strike the surface. Adsorption distorts bonds in the gas molecules or may even break them completely. This allows a reaction to proceed more easily than it would if the solid were absent.

These solid surfaces provide a new way for the reaction to occur (a new reaction pathway) that has a significantly lower activation energy.

A powdered or sponge-like form of a solid catalyst is often used to provide the greatest possible surface area. With a larger surface area, more reactant molecules can be adsorbed and the reaction is even faster.

Catalysts enable products to be made very rapidly at significantly lower temperatures than would otherwise be needed. Operating at lower temperatures reduces the energy requirements for the process. Every year, millions of tons of chemicals are produced using processes that include the most cost-effective catalysts available. Some examples of these products are shown in Table 19.1.1.

i **Adsorption** is the attraction and binding of molecules or particles of one substance to the surface of another.

TABLE 19.1.1 Some common products and the catalysts used in their manufacture

Catalyst	Reactants	Product
titanium/aluminium compound	ethene	poly(ethene)
iron	nitrogen, hydrogen	ammonia
nickel	edible vegetable oils, hydrogen	margarine
vanadium(v) oxide	sulfur dioxide, oxygen	sulfuric acid
platinum/rhodium	ammonia, oxygen	nitric acid

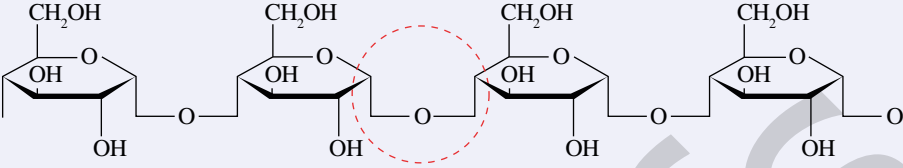
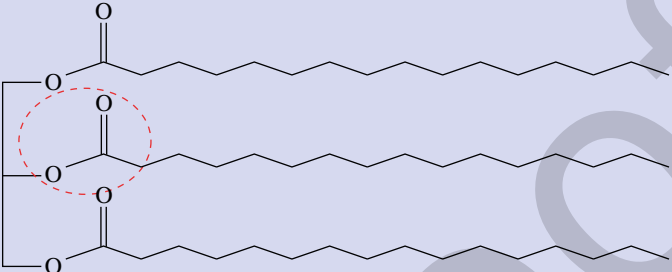
ENZYMES

Thousands of chemical reactions are involved in sustaining living things. In the same way that most industrial processes involve catalysts, so do almost all chemical reactions happening in our bodies. These biological catalysts are called enzymes.

Enzymes are highly efficient catalysts that can increase reaction rates by as much as a factor of 10^{10} . This is like taking 1 second to complete a task that normally takes 300 years.

Digestive enzymes catalyse the breakdown of large food molecules into smaller molecules. Two enzymes involved in the digestion of a potato chip are shown in Table 19.1.2.

TABLE 19.1.2 Some digestive enzymes and their role

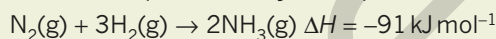
Enzyme	Role	Type of bond broken (shown circled)
amylase	breaks the ether link of starch molecules (in the mouth)	
lipase	breaks the ester link of molecules of fats and oils (in the small intestine)	

EXTENSION

Heterogeneous catalysis and the Haber process

The Haber process is a very important commercial reaction that produces ammonia gas, which is used to make fertilisers, nylon, explosives and some pharmaceuticals. In the Haber process, hydrogen and nitrogen gases are converted to ammonia (NH_3), using a catalyst of powdered iron.

The reaction is represented by the equation:



Hydrogen and nitrogen molecules both adsorb onto the iron surface (Figure 19.1.6a). As they attach themselves to the surface, the covalent bonds within their molecules break (Figure 19.1.6b).

The hydrogen and nitrogen atoms now readily combine to form ammonia molecules and move away from the iron surface (Figure 19.1.6c). The catalyst remains unaltered by the reaction.

Without a catalyst, temperatures over 3000°C are needed for a significant reaction to occur. The catalyst allows the manufacture of ammonia to proceed at an economical rate using a temperature of about 500°C .

The iron catalyst provides an alternative reaction pathway that dramatically reduces the activation energy 'barrier'—the energy needed to break the covalent bonds in the nitrogen and hydrogen molecules. Even though collisions are less frequent at 500°C than at 3000°C , a greater proportion of colliding particles have sufficient energy to successfully react.

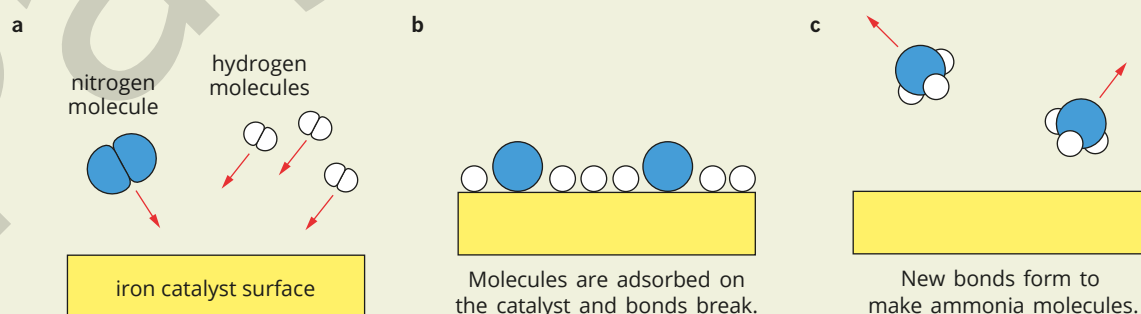


FIGURE 19.1.6 Behaviour of a catalyst in ammonia production. (a) Nitrogen and hydrogen molecules approach the iron catalyst surface. (b) The nitrogen and hydrogen molecules adsorb on the surface of the catalyst and their covalent bonds are broken. (c) The hydrogen and nitrogen atoms readily combine to form ammonia molecules. The molecules then leave the catalyst surface, and the catalyst remains unaltered by the reaction.

Enzymes are very large organic molecules, called proteins, and have specifically shaped sections that are described as active sites. These active sites can interact with particular reactants.

Figure 19.1.7 depicts the reaction pathway for an enzyme catalysing the breakdown of a reactant (the substrate) into two smaller molecules. The specific part of the enzyme molecule with which the reactant can interact is known as the **active site**.

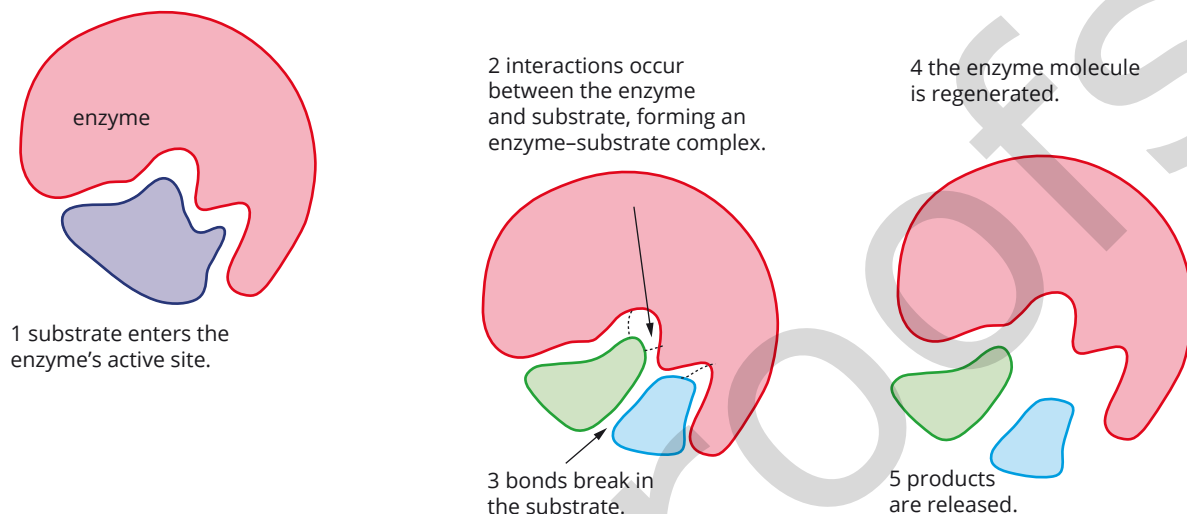


FIGURE 19.1.7 Steps in the action of an enzyme

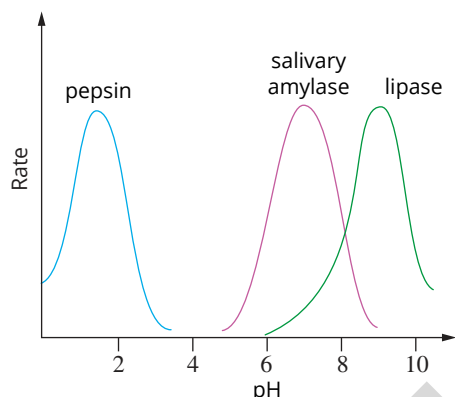


FIGURE 19.1.8 Enzymes can only operate effectively within a narrow pH range. Pepsin is a protein-digesting enzyme secreted into the stomach. Salivary amylase is the enzyme in human saliva. Lipase is the enzyme that breaks down fats. These enzymes are most effective at different pH values.

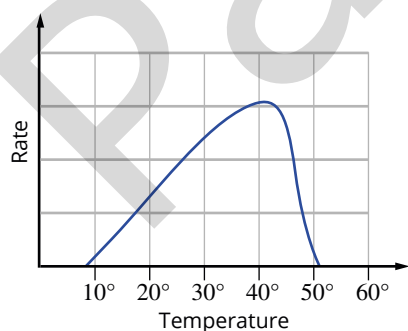


FIGURE 19.1.9 The effect of temperature on the action of an enzyme. Enzymes are only effective in a narrow range of temperatures.

Compared with inorganic catalysts, enzymes:

- can produce much faster reaction rates
- are considerably more selective, often catalysing only one specific biochemical reaction. An inorganic catalyst, such as platinum, can be used to catalyse many different reactions, often using a variety of reactants.
- are much more sensitive to pH changes and temperature changes because the intricate three-dimensional structure of their active site is destroyed at high temperatures and altered as pH changes.

Figure 19.1.8 shows the narrow pH ranges in which some common enzymes work and Figure 19.1.9 shows the effect of temperature on enzyme action.

Enzymes have numerous domestic and industrial applications. Some detergents contain enzymes that enable protein stains or greasy fat residues to be removed effectively. Fabric softeners may contain enzymes to reduce pilling of cotton fabrics.

The manufacture of vinegar, wine, beer and bread all take advantage of enzymes present in yeast to catalyse reactions such as the fermentation of sugar to ethanol:



The production of yoghurt takes advantage of enzymes present in bacteria that convert lactose, the sugar in milk, to lactic acid.

Enzymes are also used during paper manufacturing and recycling processes. They are used in the textile industry to 'stonewash' denim jeans and in the leather industry to make softer leather. In the future, when fossil fuel supplies are likely to be limited, enzymes may play an important role in the biofuel industry. For example, one group of chemists is researching the feasibility of using plant enzymes to catalyse the production of commercial quantities of ethene from biomass.

NANOPARTICLES AS CATALYSTS

Since catalysis often involves a reaction of gases passing across a metal or metal oxide surface, it is not surprising that chemists try to optimise the total surface area of a solid catalyst. Metal nanoparticles, which have a very large surface area, are therefore an exciting prospect for catalyst development.

With their much greater surface area, contact between reactant particles and catalyst is increased which means more catalytic reactions can occur in any given time. This makes the reaction occur even faster.

Considerable research effort around the world is currently devoted to identifying the best size for nanocatalysts, the most suitable design and support materials for them, and the variety of reactions they can be applied to.

One car manufacturer recently announced a new pollution-control catalyst that features nano-sized metal particles embedded in fixed positions on a ceramic base, as seen in Figure 19.1.10. The innovation is claimed to use 70% less metal and to overcome the problem of small metal particles sticking together when heated (agglomeration), which lowers the effectiveness of the catalyst.

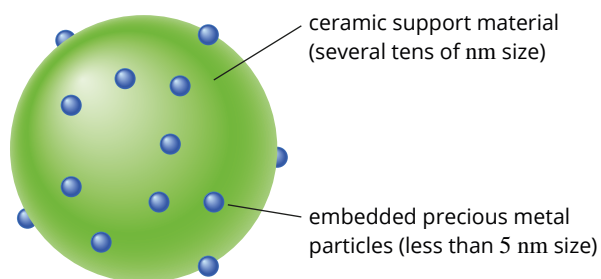


FIGURE 19.1.10 Catalyst nanoparticles on the surface of a ceramic support improve the efficiency of pollution control systems in cars.

Another research group, also focused on pollution prevention, has designed a ‘nanocage’ for a catalyst to remove unwanted sulfur from syngas fuel—a fuel that is produced in the petrochemical industry. This nanocage uses a cage-like 3D structure made of silica to protect nickel and copper nanocatalysts from heat damage.

CHEMISTRY IN ACTION

Small size, big potential

Fuel cells, like batteries, convert chemical energy into electrical energy but unlike batteries, where all the reactants are eventually reacted to make the electrical energy, fuel cells have a continuous supply of reactants to provide the electrical energy. Hydrogen fuel cells use hydrogen gas or methanol to provide hydrogen ions which react with oxygen to produce water. Catalysts speed the production of hydrogen ions, and the breaking of the bond in oxygen molecules, so the reaction occurs at rates that make the hydrogen fuel cells practical. Platinum metal, which is very expensive, is the catalyst most often used for these reactions. Platinum catalysts are now being used in the form of nanoparticles, which means less metal is required as well as giving high catalytic activity. This is contributing to lower costs for fuel cells. Other nano-based catalysts are incorporating less expensive metals to further reduce the need for expensive platinum.

Researchers have made nanowires with a nickel oxide core surrounded by a platinum shell. The nickel was leached out leaving a rough surface with increased

surface area. The catalytic activity shown is one of the highest reported.

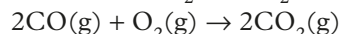
The reaction involving oxygen is very slow due to the strong double bond in an oxygen molecule. A nanoplate catalyst with a platinum/lead core surrounded by a platinum shell has shown high catalytic activity for this oxygen reaction. As well as improved catalytic activity, these nanoplate catalysts have better durability. In the acid environment of the fuel cell, in other shell-structure catalysts acid leaches through the thin outer shell to react with the metal in the core reducing catalyst stability and its lifetime. This new nanoplate catalyst has shown improved durability.

One of the requirements for the operation of a fuel cell is a semipermeable membrane that allows only hydrogen ions to pass through but no other atoms such as oxygen. Although not giving any catalytic effect, nanomaterials are also being used to improve the semipermeable membrane, allowing for lighter weight and more efficient fuel cells.

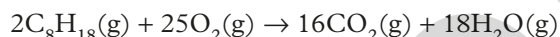
CATALYTIC CONVERTERS IN CARS

All new cars sold in Australia have a catalyst fitted between the engine and the exhaust pipe. The purpose of the catalyst is to clean the exhaust gases from the engine and reduce the air pollution that could be caused if these gases entered the atmosphere.

Catalysts in cars convert carbon monoxide and nitrogen oxide, formed in the engine, to the non-toxic gases carbon dioxide and nitrogen. Several reactions are involved in this process, including:



Unburnt hydrocarbons are also converted by the catalyst to carbon dioxide and water:



The catalyst is usually a mixture of platinum, palladium and rhodium metals and aluminium oxide, and it is mounted on a honeycomb-shaped support (Figure 19.1.11). Millions of tiny pores in the metals provide a large surface area.

Exhaust gases enter the catalyst chamber, pass quickly over the metals, and leave the exhaust pipe in a purified condition. The catalyst is unchanged by the reaction and can be used without replacement for many years.

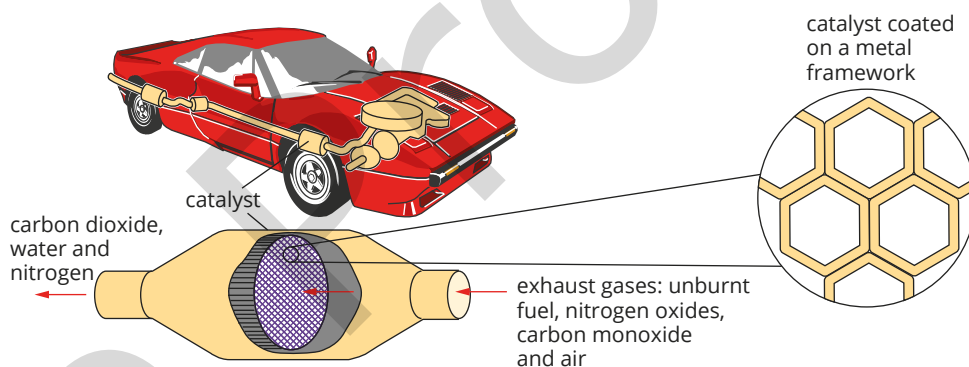


FIGURE 19.1.11 A catalyst fitted in the exhaust system of a car reduces air pollution.

Catalytic converters could not be used until unleaded petrol was available in Australia. Unleaded petrol was introduced in Australia in 1985 with the introduction of catalytic converters following in 1986 (since 2002 all petrol in Australia is unleaded). Lead, in the form of tetraethyl lead, was previously added to petrol to make engines run more smoothly and improve fuel efficiency. Catalytic converters cannot be used with leaded petrol because lead in exhaust gases would coat the surface of the metal catalysts and this would prevent the other components in exhaust gases coming into contact with the catalyst, thus making the catalyst ineffective. The phasing out of leaded petrol had a two-fold advantage in terms of air quality. Lead, like many heavy metals, is a neurotoxin (harmful to the nervous system), particularly in children, so its removal from car exhaust gases has reduced health problems. As well, its removal from petrol allowed the use of catalytic converters which reduce the concentration of the toxic gases carbon monoxide and nitrogen oxides coming from car exhausts.

19.1 Review

SUMMARY

- A catalyst provides an alternative reaction pathway that has a lower activation energy.
- Energy profile diagrams (Figure 19.1.12), which can include catalysed and uncatalysed pathways, may be used to represent the enthalpy changes and activation energy associated with a chemical reaction.

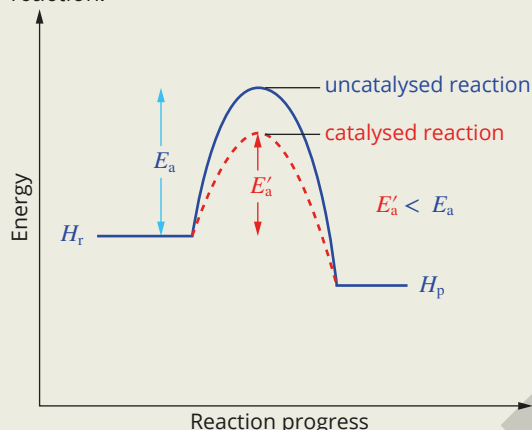
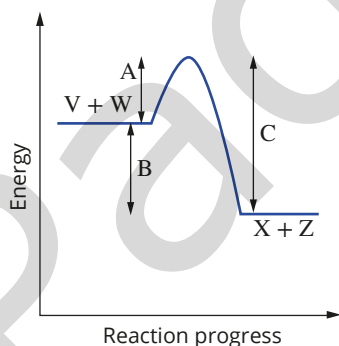


FIGURE 19.1.12 Energy profile diagram of a reaction showing the effect of a catalyst

- A catalyst provides a new reaction pathway and it is not used up in the reaction.
- When a catalyst is present, a greater proportion of the collisions between particles exceed the activation energy barrier of the reaction and therefore lead to a chemical change.
- Homogeneous catalysts are in the same state as the reactants and products of the reaction, whereas heterogeneous catalysts are in a different physical state from the reactants and products of the reaction.
- Catalysts are important in industry to enable faster production of chemicals, and to lower production costs.
- Enzymes are proteins that act as catalysts in biological systems.
- Metal nanoparticles have very large surface area to mass ratios and can be cost-efficient catalysts for commercial applications.
- Catalytic converters are used to remove pollutant gases from automobile exhausts.

KEY QUESTIONS

- 1** Consider the reaction between solutions V and W that produces X and Z according to the equation:
- $$V(aq) + W(aq) \rightarrow X(aq) + Z(aq)$$
- The energy profile diagram for this process is shown in the figure below.



Which one of the following alternatives describes the change that a catalyst produces to increase the reaction rate?

- A** B only is decreased.
B A only is decreased.
C A, B and C are decreased.
D A and C only are decreased.
- 2** Explain the meaning of:
- a catalyst
 - activation energy.
- 3** If a sugar cube is held in the flame of a candle, the sugar melts and browns but does not burn. However, the cube burns if salt is first rubbed into it, even though the salt does not react. Explain the effect of the salt on the activation energy of this combustion reaction.
- 4**
- Explain why surface properties are important to the operation of catalysts.
 - Many industrial catalysts are made into porous pellets. What is the reason for this?

Chapter review

KEY TERMS

active site
catalysis

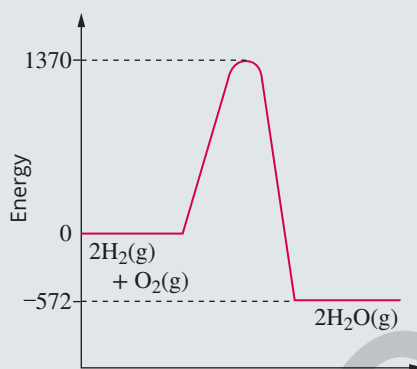
heterogeneous catalyst
homogeneous catalyst

reaction pathway

Catalysts

- The Haber process involves the reaction of nitrogen gas and hydrogen gas to make ammonia gas. Describe two ways the rate of this reaction could be increased, at constant temperature. Using collision theory, explain why rate is increased.
- Many major car makers have plans for hydrogen-powered cars. In the fuel cells of these cars, hydrogen reacts with oxygen from the air to produce water.

$$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$$
 Energy changes for the reaction are shown in the graph below.



- What is the magnitude of the activation energy of this reaction?
 - What is ΔH for this reaction?
 Several groups of scientists have claimed to have split water into hydrogen and oxygen using a molybdenum catalyst:

$$2\text{H}_2\text{O}(\text{g}) \xrightarrow{\text{Mo}} 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$$
 - Sketch energy profile diagrams for this reaction with and without the presence of a catalyst.
 - What is the value of ΔH for this water-splitting equation?
- For the reactions a-d below, state whether the catalyst is homogeneous or heterogeneous. Give a reason for your answer.
 - Hydrogen peroxide solution is decomposed to oxygen gas and water in the presence of aqueous dichromate ions.
 - Sulfur dioxide gas is reacted with oxygen gas to produce sulfur trioxide gas in the presence of solid vanadium pentoxide (V_2O_5).

- Methane (CH_4) and steam react to produce hydrogen gas and carbon monoxide gas in the presence of nickel metal.
 - Conversion of aqueous ethanol to acetic acid in the presence of an enzyme.
- Why do chemists prefer to use heterogeneous catalysts for industrial processes?
 - State whether the following statements about catalysts are true or false. Explain your answer.
 - The quantity of products from a reaction is increased when a catalyst is used.
 - Catalysts can be recovered unchanged from a reaction.
 - Catalysts increase the rate of a reaction by increasing the energy with which reactant particles collide so that a higher proportion of collisions exceed the activation energy.
 - The enthalpy of a reaction is decreased in the presence of a catalyst.
 - A finely divided nickel catalyst is selected to catalyse the hydrogenation of vegetable oils to produce margarine. Which one of the following would *not* be a reason for using this catalyst?
 - Nickel is less toxic than other metals that are more effective catalysts.
 - Finely divided nickel will be a more effective catalyst than nickel plate.
 - Nickel lowers the energy change of the reaction involved more than other metals do.
 - The finely divided nickel is readily removed from the reaction mixture.
 - The reaction of tartrate ion ($\text{C}_4\text{H}_4\text{O}_6^{2-}$) and hydrogen peroxide (H_2O_2) is shown below.

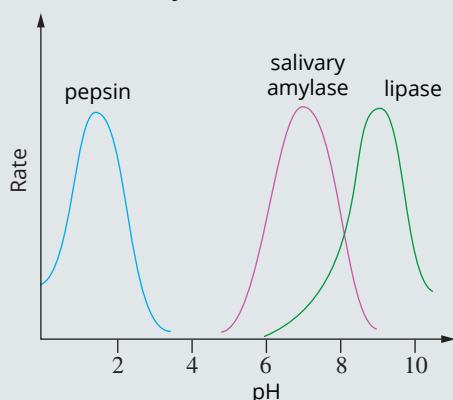
$$\text{C}_4\text{H}_4\text{O}_6^{2-}(\text{aq}) + 5\text{H}_2\text{O}_2(\text{aq}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l}) + 2\text{OH}^-(\text{aq})$$

At room temperature, little reaction occurs but addition of pink cobalt(II) ions (Co^{2+}) to the reaction mixture causes rapid effervescence and a colour change from pink to green and back to pink again. What evidence is there that the cobalt(II) ions are acting as a catalyst? Account for the colour change in your answer.

- 8 In the production of lactose-free milk, the enzyme lactase is used to catalyse the conversion of lactose into galactose and glucose. Which of the following statements about this catalysed process are correct? (More than one answer is possible.)

A Lactase is a protein.
B Lactose is described as the substrate.
C Lactose has an 'active site' where bond breaking occurs.
D Glucose is described as the substrate.

- 9 The graph below shows the range of activity for the enzymes pepsin, salivary amylase and lipase. Why are these enzymes active only over a narrow pH range? In your answer refer to the ability of the substrate to interact with the enzymes.



- 10 Describe how catalytic converters in motor vehicles are able to limit the potential for atmospheric pollution.

- 11 In the catalytic converter of a car, carbon monoxide gas is removed from exhaust gases by passing the gas mixture over a platinum, Pt, catalyst. Sort the sequence of steps W–Z below to describe the pathway, in order, for this catalysed reaction.

W Adsorbed O atoms combine with adsorbed CO molecules.

X O₂ and CO molecules adsorb on to the Pt surface.

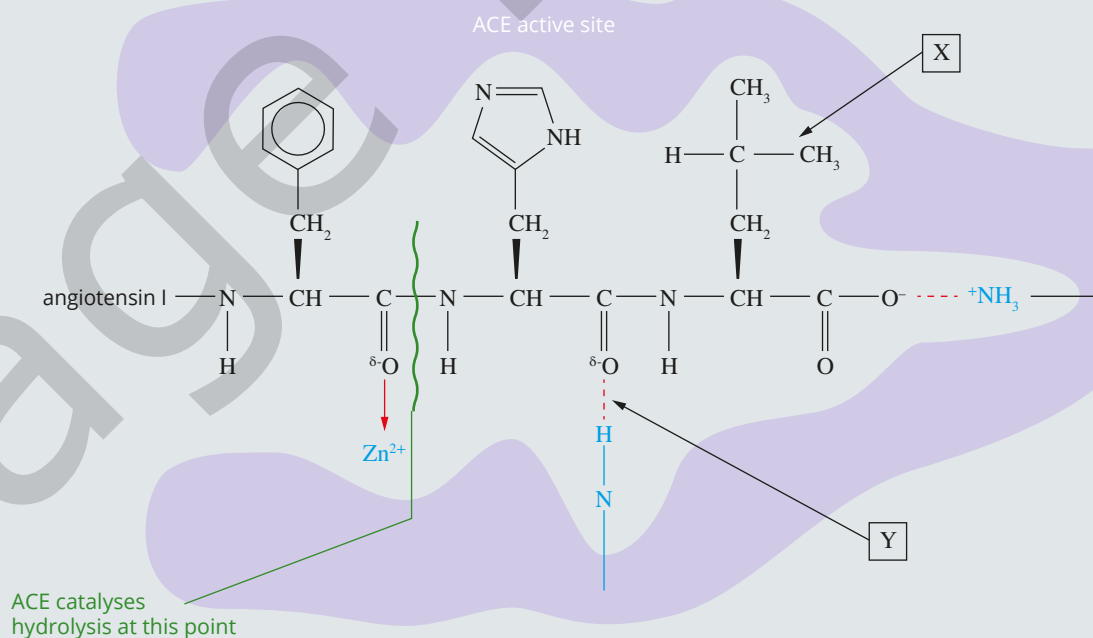
Y Adsorbed CO₂ molecules leave the Pt surface.

Z Covalent bonds within O₂ molecules break.

- 12 When a mixture of hydrogen and oxygen gases is passed over a platinum gauze, the mixture spontaneously ignites. State the role of the platinum gauze and explain how it causes the hydrogen and oxygen gases to ignite.

- 13 Give two reasons that nanomaterials are seen as having great potential as catalysts.

- 14 The figure below shows the interactions at the active site of an enzyme, called ACE, responsible for catalysing a biochemical reaction in which a small section is chopped off a short polypeptide molecule. The product of this reaction is important in controlling blood pressure.



Which one of the following statements about this enzyme-catalysed reaction is correct?

- A** ACE would be described as the substrate in this process.
B The part of the molecule labelled X interacts with the active site because it is polar.

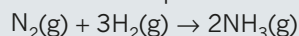
C The interaction labelled Y (the dashed line) represents a hydrogen bond.

- D** This biochemical reaction would have an even faster rate if the temperature was lowered from 37°C to 20°C.

- 15** Use the word or phrases from the following list to describe what happens to nitrogen and hydrogen molecules when their reaction to produce ammonia is catalysed at an iron surface by completing the sentences at A–G below.

absorb	dispersion	molecules
adsorb	form covalent	react with iron
atoms	bonds	strengthened
broken	hydrogen	strong
covalent	increasing	weak
decreasing	ions	

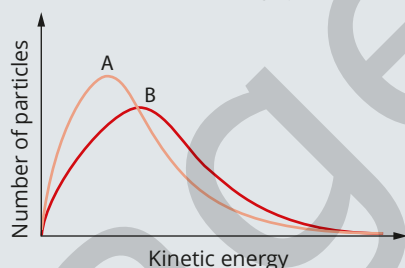
A balanced equation for the reaction is:



Nitrogen molecules, N_2 , consist of two nitrogen (A) held together by a triple covalent bond. This is a particularly (B) covalent bond and so the reaction has a large activation energy in the absence of a catalyst. Iron is a suitable catalyst for this reaction, capable of providing a new reaction pathway and so (C) the activation energy.

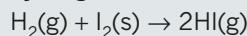
Nitrogen and hydrogen molecules (D) onto the iron surface and form bonds with the iron surface. The (E) bonds in the N_2 and H_2 molecules are (F) and individual neighbouring nitrogen and hydrogen atoms (G) and become ammonia molecules. The ammonia molecules then leave the iron surface.

- 16 a** The figure below shows the distribution of energies of particles in a substance at two different temperatures, 40°C and 60°C . Indicate the temperatures represented by graphs A and B.



- b** Copy this diagram for temperature B and use the diagram to show the effect of a catalyst on a reaction.
- c** Use the diagram you have drawn in part **b** to explain in terms of collision theory how a catalyst increases the rate of a reaction.

- 17** The graph below shows the energy profile diagram for the reaction of hydrogen and iodine to form hydrogen iodide:



- a** Copy the diagram and label the following: $\text{H}_2(\text{g})$ and $\text{I}_2(\text{s})$; $\text{HI}(\text{g})$; ΔH ; activation energy.
- b** Is the reaction endothermic or exothermic?
- c** On the diagram draw the energy profile that would result if a catalyst was used in the reaction.

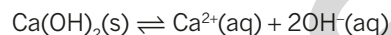
UNIT 2 • MOLECULAR INTERACTIONS AND REACTIONS

REVIEW QUESTIONS

Section 1: Multiple choice

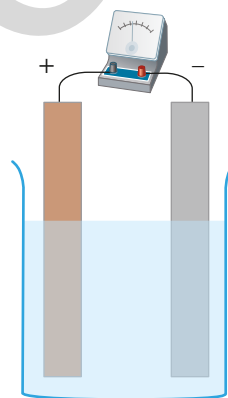
- Which of the following statements about water is correct?
 - Water molecules are non-polar and so water is a good solvent.
 - Water has a high latent heat value and so it is a good coolant.
 - Water contracts on freezing and so lakes are covered with a layer of ice in very cold climates.
 - Water has a low specific heat capacity and so large bodies of water moderate temperatures on Earth.
- Which one of the equations below best represents table sugar ($C_{12}H_{22}O_{11}$) dissolving in water?
 - $C_{12}H_{22}O_{11}(s) + H_2O(l) \rightarrow C_{12}H_{23}O_{11}^+(aq) + OH^-$
 - $C_{12}H_{22}O_{11}(s) + H_2O(l) \rightarrow C_{12}H_{21}O_{11}^-(aq) + H_3O^+(aq)$
 - $C_{12}H_{22}O_{11}(s) \xrightarrow{H_2O(l)} C_{12}H_{22}O_{11}(aq)$
 - $C_{12}H_{22}O_{11}(s) \xrightarrow{H_2O(l)} C_{12}H_{22}O_{11}(l)$
- In which one of the following reactions is the substance in bold font acting as an acid?
 - $C_6H_5COO^-$** (aq) + H_2O (aq) $\rightarrow$ OH^- (aq) + CH_3COOH (aq)
 - H_2** (g) + C_2H_4 (g) $\rightarrow$ C_2H_6 (g)
 - HSO_4^-** (aq) + PH_3 (g) $\rightarrow$ SO_4^{2-} (aq) + PH_4^+ (aq)
 - NH_3** (aq) + H_2O (l) $\rightarrow$ NH_4^+ (aq) + OH^- (aq)
- Which one of the following aqueous solutions would have the lowest pH?
 - $0.1 \text{ mol L}^{-1} H_2SO_4$
 - $0.1 \text{ mol L}^{-1} HNO_3$
 - $0.1 \text{ mol L}^{-1} H_3PO_4$
 - $0.1 \text{ mol L}^{-1} Ca(OH)_2$
- If the acidity of a solution were increased, how would the hydrogen ion concentration, $[H^+]$, and the pH be affected?
 - $[H^+]$ and the pH would increase.
 - $[H^+]$ and the pH would decrease.
 - $[H^+]$ would increase and the pH would decrease.
 - $[H^+]$ would decrease and the pH would increase.
- In which chemical reaction will an increase in pressure increase the rate of reaction?
 - $CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$
 - $H(g) + Br_2(g) \rightarrow 2HBr(g)$
 - $N_2(g) + 3F_2(g) \rightarrow 2NF_3(g)$
 - all of the above

- A saturated solution of calcium hydroxide was prepared in a flask using an excess of solid calcium hydroxide so that some remained undissolved.



A small amount of 10 mol L^{-1} hydrochloric acid was added to the flask. What would you most likely observe?

- No change is observed.
 - $CaCl_2$ is precipitated.
 - More solid $Ca(OH)_2$ is precipitated.
 - Some solid $Ca(OH)_2$ dissolves.
- When a solution of KBr is placed in a beaker connected to a power supply, a current flows through the circuit and a reaction occurs at the electrodes. The set-up is shown below.



Which one of the following statements best describes how the current is moving in the KBr solution?

- The water molecules in the solution carry the current.
 - KBr molecules move through the solution to each electrode.
 - There are freely moving ions in the solution.
 - Electrons flow through the solution to the electrodes.
- Which of the following compounds would most readily dissolve in water?
 - NH_4I
 - $AlPO_4$
 - $BaCO_3$
 - Ag_2SO_4
 - Which of the following molecules are polar?

i CH_4 ii CH_3Cl iii CH_2Cl_2 iv $CHCl_3$ v CCl_4

 - all of the above
 - none of the above
 - ii and iv
 - ii, iii and iv

UNIT 2 • REVIEW

11 The compound that would be most soluble in water is:

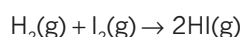
- A $\text{Al}(\text{OH})_3$.
- B CH_3OH .
- C CHF_3 .
- D $\text{CF}_2 = \text{CF}_2$.

12 If a bottle of concentrated sulfuric acid were spilt the most appropriate action would be to:

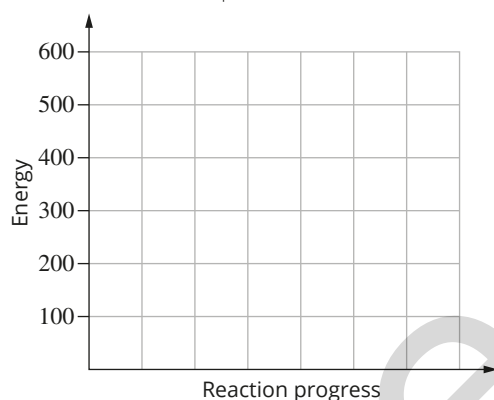
- A dilute the acid with water.
- B add sodium hydroxide to neutralise the acid.
- C cover it with powdered sodium carbonate.
- D soak it up with paper towels.

Section 2: Short answer

1 a Copy the graph below and draw the energy profile diagram for the following data for the reaction between gaseous H_2 and gaseous I_2 .



$$H_f = 250 \text{ kJ} \quad H_p = 450 \text{ kJ} \quad E_a = 300 \text{ kJ}$$



- b On your graph, show the $\Delta H_{\text{reaction}}$.
- c To catalyse the above reaction some sulfuric acid is added. The E_a of the forward reaction is lowered by 75 kJ. Draw the resulting reaction path on your graph.

2 Hydrogen peroxide, H_2O_2 , *exothermically* decomposes into water and oxygen.

- a Write an equation for this reaction, including the enthalpy as a reactant or product in the equation.
- b At room temperature, the reaction is very slow in an open beaker.

Sketch the enthalpy diagram, showing the activation energy, E_a , and the change in the enthalpy of reaction, ΔH .

- c When a catalyst is added a vigorous reaction occurs. Sketch the enthalpy diagram, labelling E_a and ΔH . Use the same scale (size) as you did for the graph in part b.

3 Write the balanced equation for each of the following reactions.

- a The molecular equation for the reaction of a sulfuric acid solution with a potassium hydroxide solution
- b The ionic equation for the reaction of a sulfuric acid solution with a potassium hydroxide solution
- c The molecular equation for the reaction of a hydrochloric acid solution with aluminium metal
- d The ionic equation for the reaction of a hydrochloric acid solution with aluminum metal.

4 a A 20.0 g sample of calcium bromide is dissolved in 400.0 mL of water.

Calculate the concentration of the solution in mol L^{-1} .

b A 1000.0 L sample of seawater is evaporated. If the concentration of sodium chloride in seawater is 0.600 mol L^{-1} , what mass of NaCl is obtained?

5 The boiling points (BP) and the molar masses (M) for the hydrides (hydrogen compounds) of the halogens are given in the following table.

Molecule	BP ($^{\circ}\text{C}$)	M (g)
HF	19	20
HCl	-85	36.5
HBr	-67	81
HI	-35	128

The boiling points increase as the molar mass increases; however, HF does not fit this trend in terms of molecular bonding. Explain. You can draw a graph to help explain your answer.

6 a In 1660 Robert Boyle observed 'that for a fixed mass of gas the volume of the gas decreases as the pressure increases provided the temperature remains constant'. (Now known as Boyle's law)

Explain Boyle's law in terms of the kinetic theory of gases.

b In 1780 Jacques Charles observed 'that for a fixed mass of gas the volume of the gas increases as the temperature increases provided the pressure remains constant'. (Now known as Charles' law)

Explain Charles' law in terms of the kinetic theory of gases.

Section 3: Extended answer

- 1** The production of the fertiliser and mining explosive, ammonium nitrate, is produced via a series of chemical reactions.

In the first stage, ammonia and oxygen from the air are heated together to produce nitrogen monoxide and water.

- a** For every 1.00 tonne (1.00×10^6 g) of ammonia calculate the:
- i** mass of oxygen required
 - ii** volume of nitrogen monoxide (NO) produced at STP.

The reaction is extremely slow and uneconomical at room temperature and pressure, therefore conditions of high concentrations of reactants at temperatures of 800°C and 10 times atmospheric pressure in the presence of a finely divided platinum catalyst, are required to increase the rate of reaction.

- b** Discuss using collision theory how each of the following conditions increases the rate of this reaction.
- i** high concentrations of the reactants ($\text{NH}_3(\text{g})$ and $\text{O}_2(\text{g})$)
 - ii** high temperature (800°C)
 - iii** high pressure (10 times atmospheric pressure)
 - iv** in the presence of a catalyst (finely divided platinum)

- 2** The solubility of compounds in water is determined experimentally and can be measured in grams of solute per 100 g of water.

The solubility of salt (sodium chloride, NaCl) in water is 36 g at 25°C .

- a** Based on the data above, what other way could the solubility can be expressed?
- b** Calculate the solubility of NaCl expressed as moles per 100 g of solution.
- c** Calculate is the solubility of NaCl in moles per litre (mol L^{-1})? (Assume the mass of 1.0 L is 1000.0 g.)

15.0 g of NaCl was added to 25 g of water in a beaker at 25°C , stirred vigorously for a few minutes then left to stand.

- d** Would the solution be unsaturated or saturated?
- e** Why is the temperature given 'at 25°C ' when reporting the solubility?
- f** Describe the appearance of the contents of the beaker. You may include a diagram in your answer.
- g** Based on your understanding of the properties of ionic solids, explain why a solid crystal of NaCl is a non-conductor of electricity, however, when dissolved in water, the aqueous solution is a conductor of electricity. Include an equation in your answer.
- h** The solubility of sugar (sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is 68.0 g per 100.0 g of water at 25°C .

Although sugar has a higher solubility in water than sodium chloride, the solution has almost zero electrical conductivity. Based on your understanding of the properties of materials, explain the difference between the conductivity of a salt solution and a sugar solution.

Page Proofs